Data Science Fundamentals

1st Week – 2nd session

2024/25

Python – What is it?

Object-oriented programming language (just like Java, C++, etc.)

Interpreted language (different than compiled languages – Java, C++)

Executes code line-by-line (also different than compiled languages)

Dynamically typed: During execution, the data type is checked and inferred automatically (quite different from Java, C, etc.)

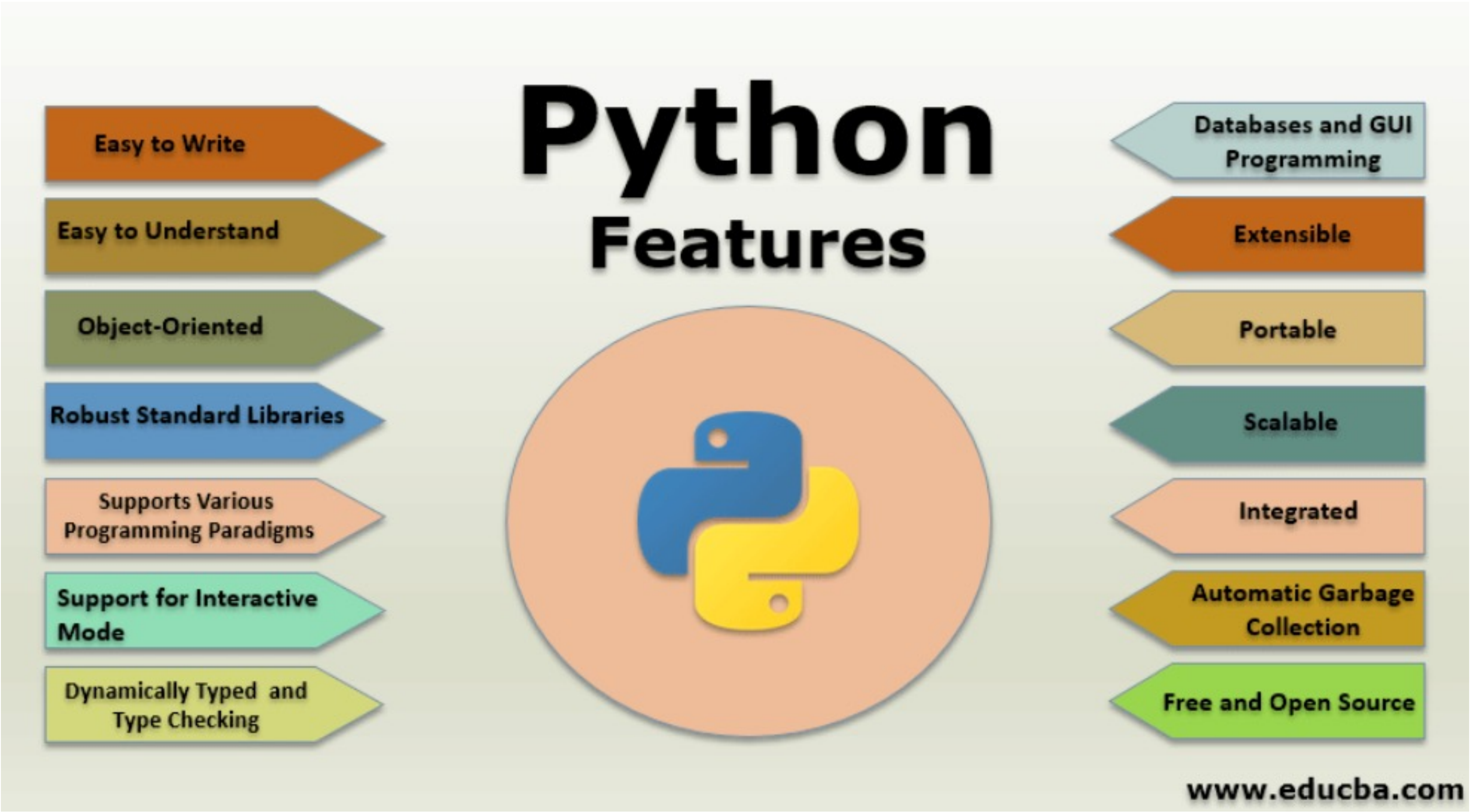
Python – A bit of history

Created in the late 80s by Guido van Rossum

Named after the famous BBC television series – Monty Python.

It was the successor of the ABC programming language, as it was able to handle exceptions and was integrated into the Amoeba operating system.

Python – Why use it?



Python – Advantages & Disadvantages

Advantages

- Excellent for data exploration (and thus, for data science)
- A large collection of powerful libraries that are free to use and simple to install (pip/conda install package name).
- Very active scientific and academic community!
- Write Once, Run Everywhere!, just like Java

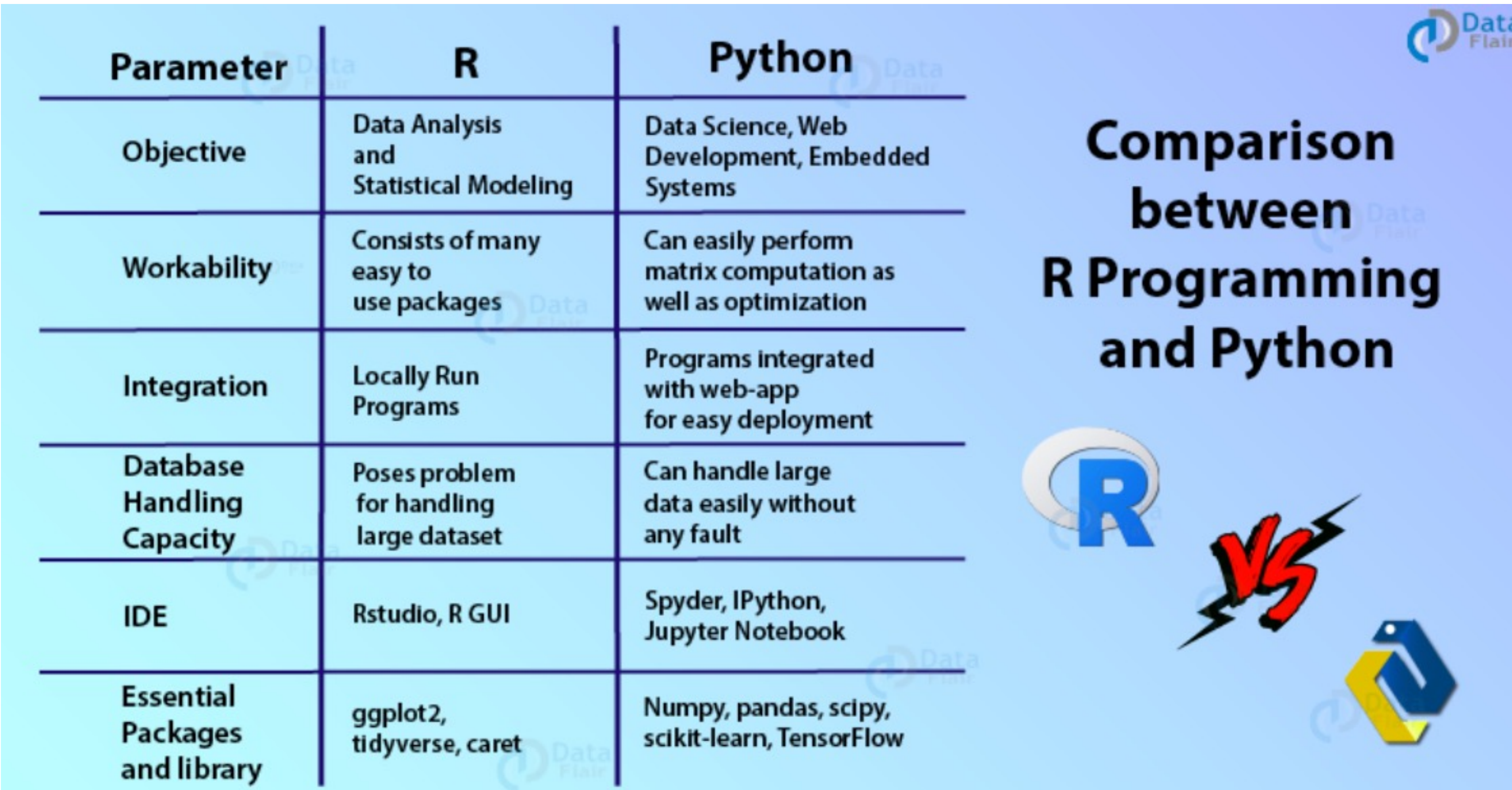
Disadvantages

- Slower (because it's interpreted line by line – but makes debugging much easier!)
- Sometimes, it is inefficient in memory consumption
- Weak in mobile application development

Python – A universe of applications

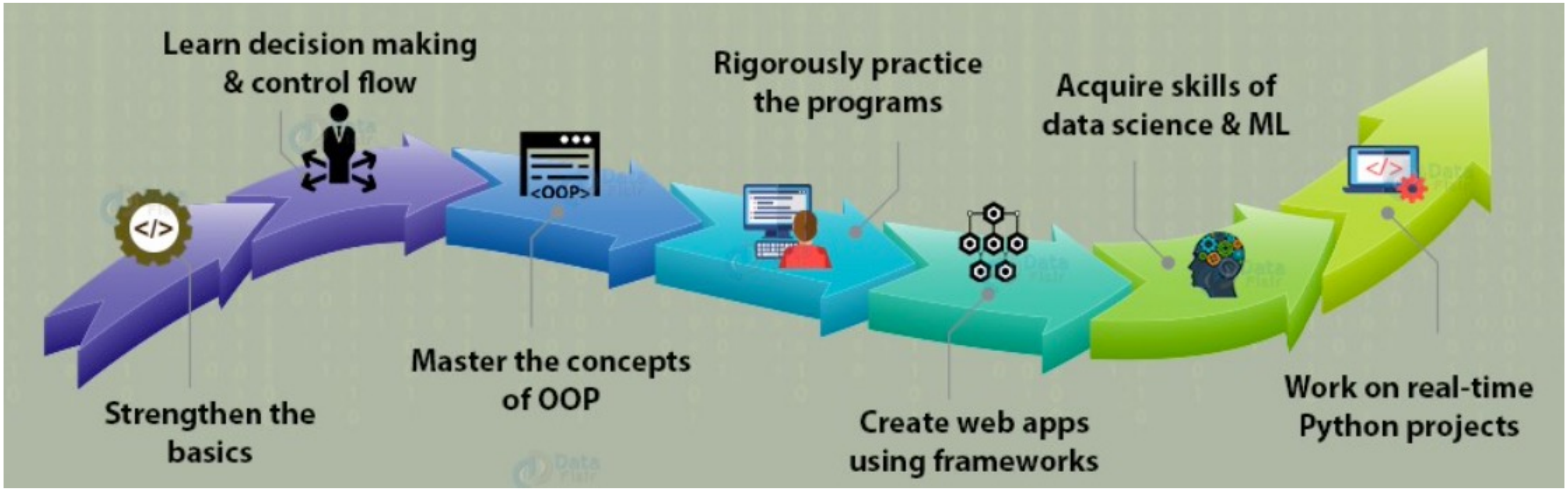
Python – Some Python libraries for Data Science

Python – Why Python for Data Science instead of R?



Parameter	R	Python
Objective	Data Analysis and Statistical Modeling	Data Science, Web Development, Embedded Systems
Workability	Consists of many easy to use packages	Can easily perform matrix computation as well as optimization
Integration	Locally Run Programs	Programs integrated with web-app for easy deployment
Database Handling Capacity	Poses problem for handling large dataset	Can handle large data easily without any fault
IDE	Rstudio, R GUI	Spyder, IPython, Jupyter Notebook
Essential Packages and library	ggplot2, tidyverse, caret	Numpy, pandas, scipy, scikit-learn, TensorFlow

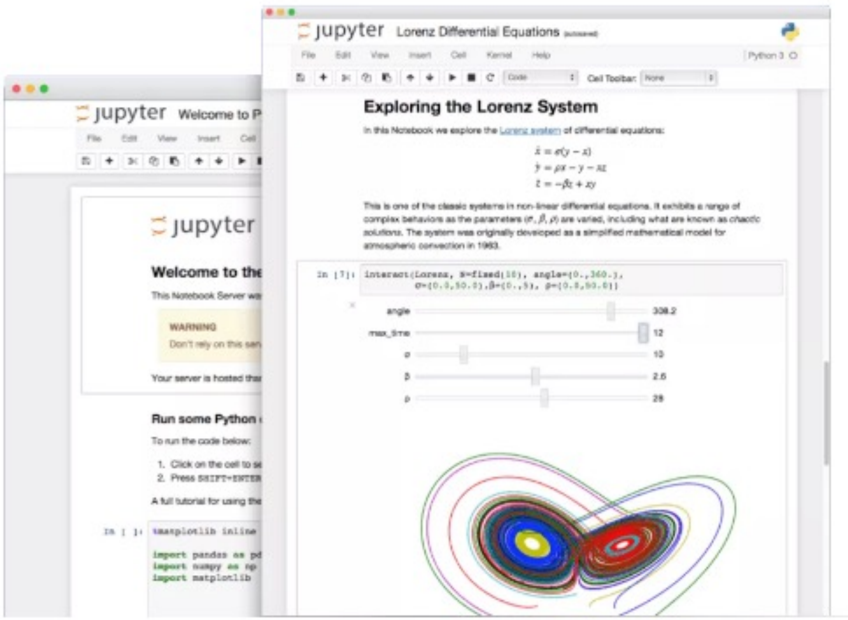
Python – Learning flow in Data Science



JupyterLab – What is it?

- Inspired by the scientist's notebook
- Open-source platform for exploratory and interactive computing
- IPython Notebook 2011
- Jupyter Notebook 2014
- Supports a variety of programming languages
- Jupyter is an acronym for Julia (Ju), Python (Py), and R
- JupyterLab 2018, 2019, 2021–22

https://jupyter.org/



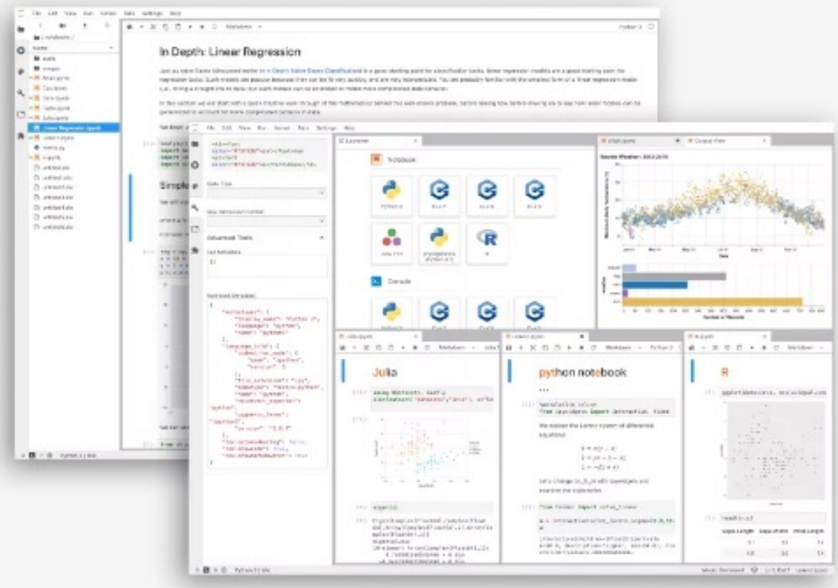
Jupyter Notebook: The Classic Notebook Interface

The Jupyter Notebook is the original web application for creating and sharing computational documents. It offers a simple, streamlined, document-centric experience.

Try it in your browser

Install the Notebook

https://jupyter.org/



JupyterLab: A Next-Generation Notebook Interface

JupyterLab is the latest web-based interactive development environment for notebooks, code, and data. Its flexible interface allows users to configure and arrange workflows in data science, scientific computing, computational journalism, and machine learning. A modular design invites extensions to expand and enrich functionality.

Try it in your browser | Install JupyterLab

Let's try in our browser...

JupyterLab

The latest web-based interactive development environment

Jupyter Notebook

The original web application for creating and sharing computational documents

JupyterLite

JupyterLite (Wasm powered Jupyter) deployed as static GitHub Pages

Jupyter Widgets

HTML widgets in Jupyter notebooks for interactive exploration of input data

Voilà

Share insights by converting notebooks into interactive dashboards

JupyterLab Welcome Tour

Help -> Welcome Tour

JupyterLab documentation

https://jupyterlab.readthedocs.io/en/stable/

From JupyterLab documentation

- A computational notebook is a shareable document that combines computer code, plain language descriptions, data, rich visualizations like 3D models, charts, graphs and figures, and interactive controls.

- A notebook, along with an editor like JupyterLab, provides a fast interactive environment for prototyping and explaining code, exploring and visualizing data, and sharing ideas with others.

JupyterLab installation

- JupyterLab can:
  - run online – https://jupyter.org/try-jupyter/lab/index.html

    or
  - can be installed and run in your computer
    - 1st install Python (for using pip)
    - then execute in a terminal: *pip install jupyterlab*
    - and then run in a terminal: *jupyter lab*

both versions will run on your browser

- Supported browsers:
  - Firefox
  - Chrome
  - Safari

How to Use JupyterLab

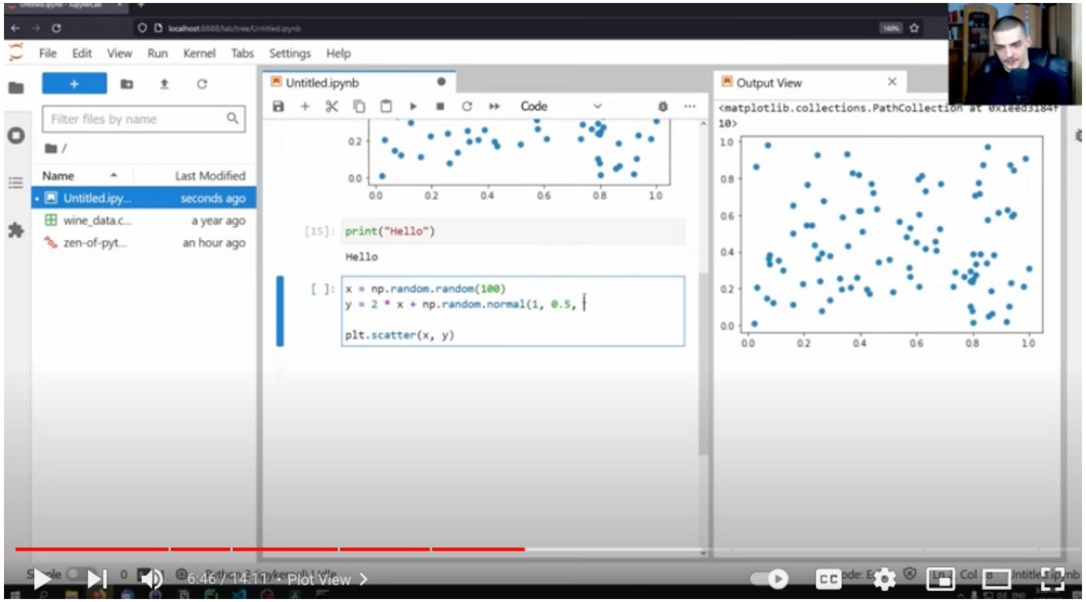
an introduction to JupyterLab, Markdown and Python on JupyterLab, on YouTube

https://www.youtube.com/watch?v=A5YyoCKxEOU&t=294s

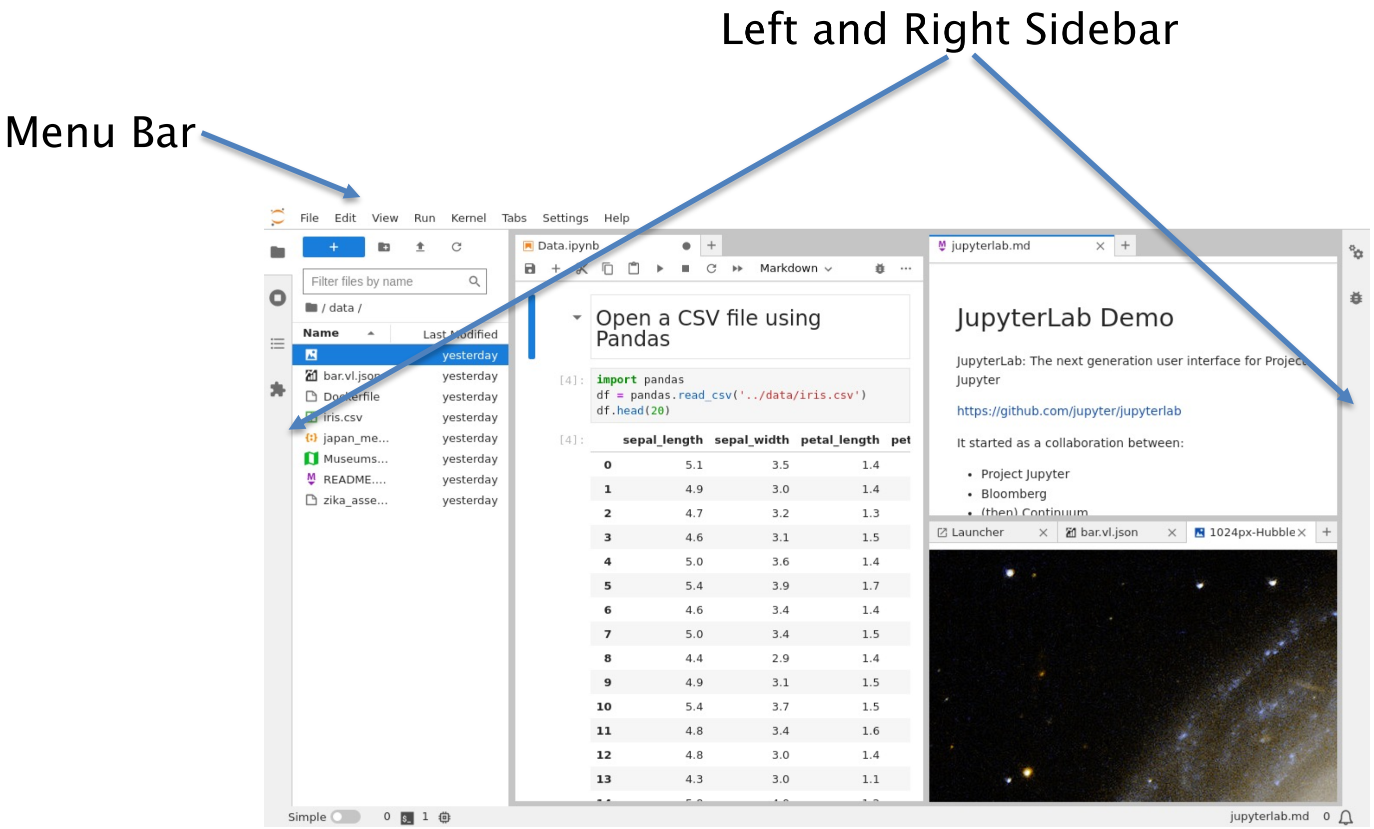
Jupyter Lab is AWESOME For Data Science

also on YouTube, more advanced on how to use Python on notebooks

https://www.youtube.com/watch?v=7wf1HhYQiDg

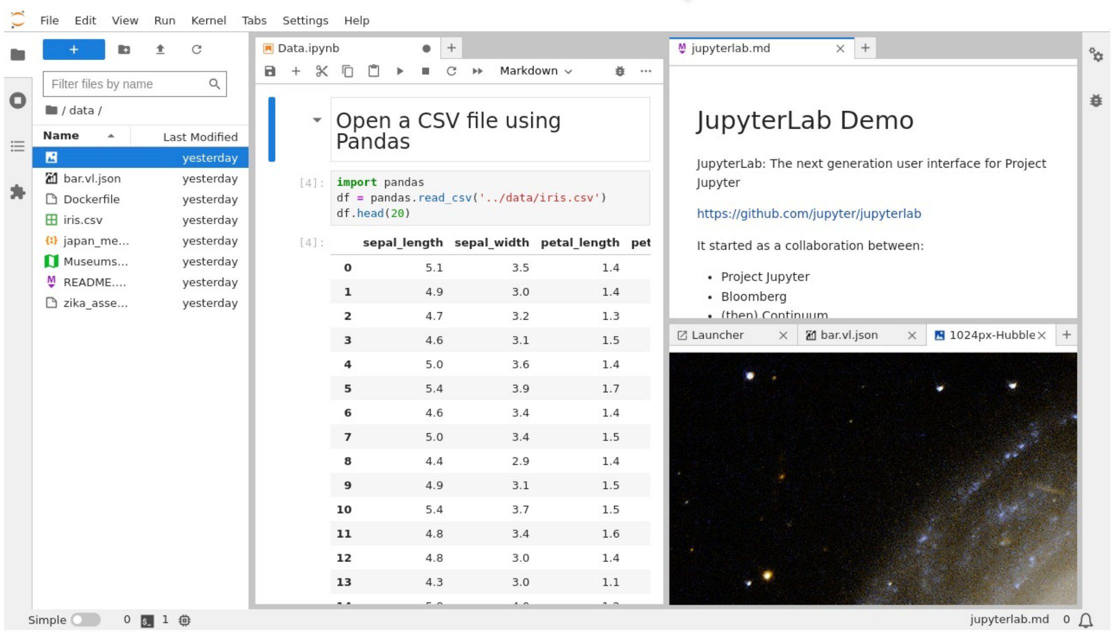


The JupyterLab Interface



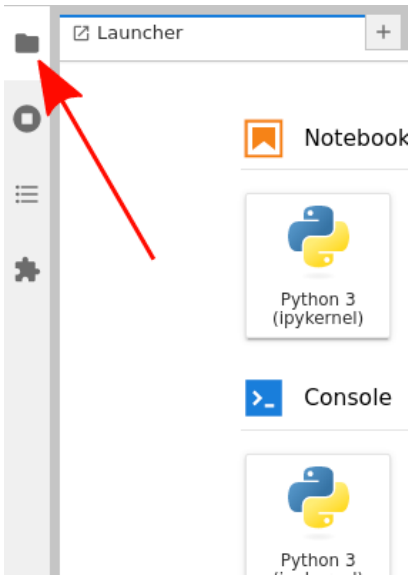
The JupyterLab Interface

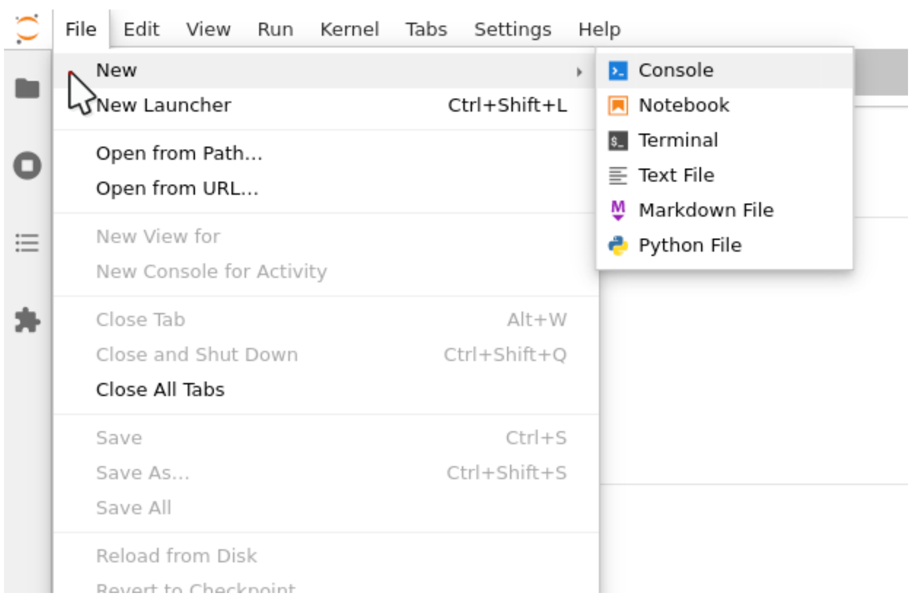
Main Work Area



Context Menus can be accessed by right-clicking on each element

Creating/opening/managing Files

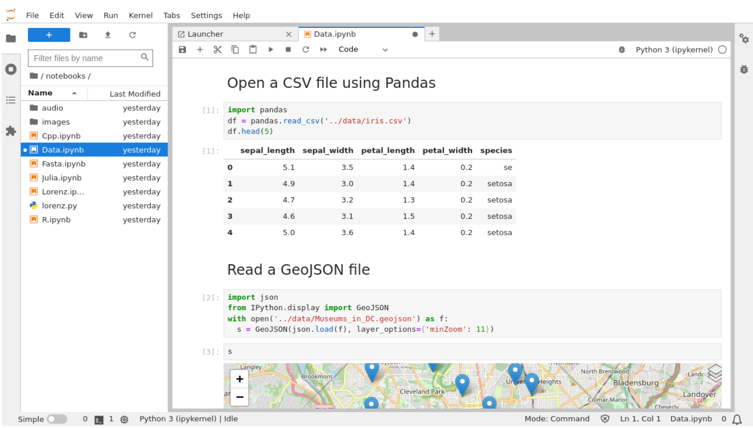


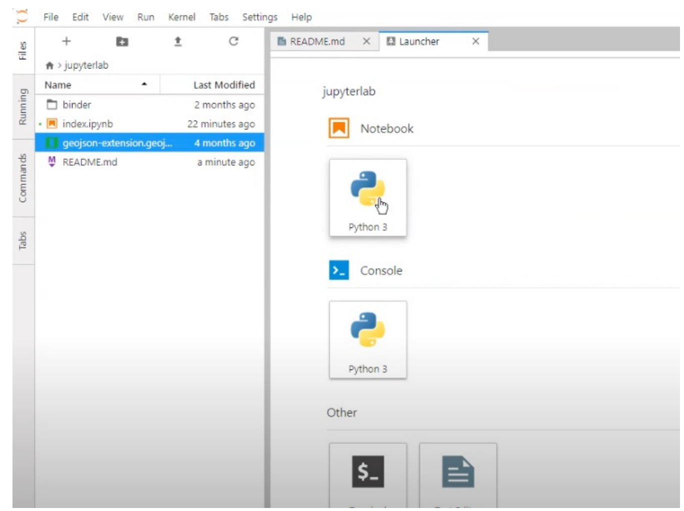


also, you can drag-and-drop files to the main work area

Notebooks

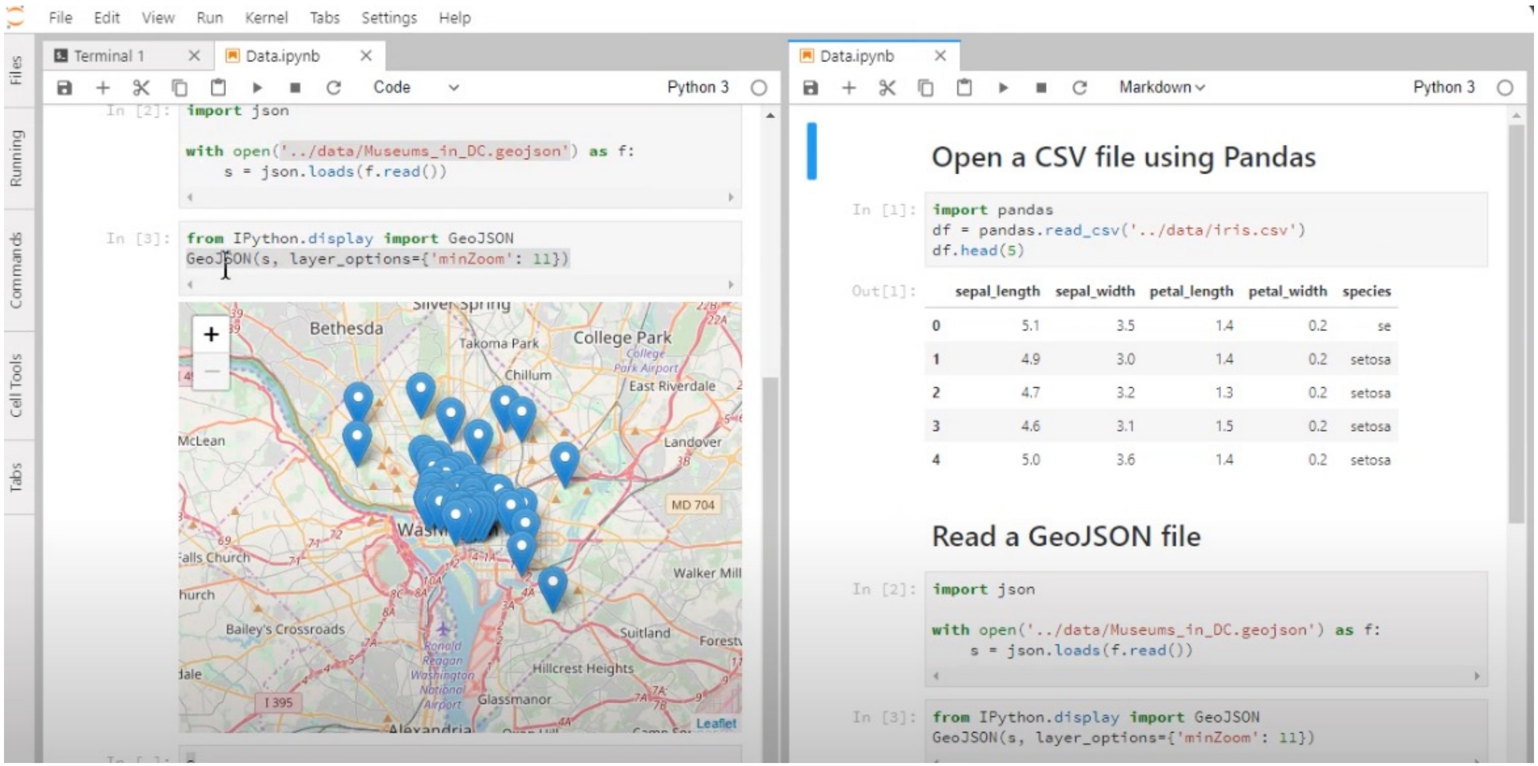
Jupyter notebooks are documents that combine live runnable code with narrative text (Markdown), equations (LaTeX), images, interactive visualizations and other rich output





Create a notebook by clicking the + button in the file browser and then selecting the Python kernel in the new Launcher tab

Create multiple synchronized views of a single notebook



Markdown in a JupyterLab notebook

Search for "Markdown in Jupyter Notebook Tutorial"

Markdown in a JupyterLab notebook

Headings

(Header 1, title)
(Header 2, major headings)
(Header 3, subheadings)
(Header 4)
(Header 5)
(Header 6)

<h1>Header 1,title<h1>
<h2>Header 2,major headings<h2>
<h3>Header 3,subheadings<h3>
<h4>Header 4<h4>
<h5>Header 5<h5>
<h6>Header 6<h6>

Markdown in a JupyterLab notebook

Blockquotes

>This is good

<blockquote>This is good</blockquote>

Markdown in a JupyterLab notebook

Code Section

Using Markdown

````
```Python
str = "This is block level code"
print(str)
```
````

Using Markup Tags

```
<code>Python
str = "This is a block level
code"
print(str)
</code>
```

Markdown in a JupyterLab notebook

Line Break

Bold Text

This is bold text

** This is bold text

__ This is bold text

Markdown in a JupyterLab notebook

Italic text

<i>This is italic text </i>

* This is italic text

_ This is italic text

Markdown in a JupyterLab notebook

Horizontal Line

Ordered List

ol>
Fish
Eggs
Cheese

1. Fish
2. Eggs
3. Cheese

Markdown in a JupyterLab notebook

Unordered List

Fish
Eggs
Cheese

- Fish
- Eggs
- Cheese

Markdown in a JupyterLab notebook

Internal Link

Define a section with an ID:

    [Section title](#division_ID)

Then you can make an internal link:

    go to Section title

The result would be the link

go to Section title

Markdown in a JupyterLab notebook

External Link

<a> href="https://www.google.com" >Link to Google

Markdown in a JupyterLab notebook

Table

```
<table style="width:20%">
  <tr>
    <th><Name></th>
    <th><Address></th>
    <th><Salary></th>
  </tr>
  <tr>
    <td><Hanna></td>
    <td><Brisbane></td>
    <td><4000></td>
  </tr>
</table>
```

Markdown in a JupyterLab notebook

Images

```
<img src="https://i.imgur.com/WWrydEh.png"
                                   width ="500" height=500 >
```